

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of BSc Biological Sciences Examination November 2019

Course code & Name: BS309 Molecular Biology

06.11.2019 Wednesday 10:00 AM TO 10:30 AM Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
 - Use of non-programmable calculator is allowed.
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Q - I Choose the correct answer for the following questions.**20**

1. Select the wrong combination
(i) Griffith/ pneumonia/ transformation (ii) MacLeod/ *D pneumoniae* / transformation
(iii) Avery / *D pneumoniae* / P³² (iv) Chase/ Virus/ P³²
2. Which of the following is a DNA virus
(i) Bacteriophage lambda (ii) HIV
(iii) Ebola virus (iv) Rabies virus
3. Chromatin is a complex of
(i) RNA and Proteins (ii) DNA and Proteins
(iii) DNA and RNA (iv) None of the above
4. C-value paradox is related to
(i) genome size and organism complexity (ii) cell size and organism complexity
(iii) proteome size and organism complexity (iv) transcriptome size and organism complexity
5. Dystrophin gene mutation can be associated with the disease related to
(i) Bones (ii) Liver (iii) Muscles (iv) Kidney
6. 2019 Nobel prize in Physiology and Medicine was awarded for the discovery of
(i) how cells sense and adapt to oxygen availability (ii) cancer immune therapy
(iii) molecular mechanisms controlling the circadian rhythm (iv) mechanisms for autophagy
7. To initiate replication which of the following mechanism requires a “nick”
(i) Rolling circle (ii) Strand displacement
(iii) Theta (iv) All the above requires a nick
8. Full form of LINEs is
(i) Long Interspread Nuclear Elements (ii) Long Interspersed Nuclear Elements
(iii) Long Interspread Nucleus Elements (iv) Long Interspersed Nucleus Elements
9. Instrument used to detect and measure ionizing radiation is called
(i) Hershey-Chase Counter (ii) IonoRadio Counter
(iii) Geiger–Muller Counter (iv) Radiophoto Counter
10. Select the wrong combination
(i) Tus/ Replication (ii) Holliday Junction/ Recombination
(iii) Cyclin/ Cell division (iv) Branch point/ Transcription

11. NHEJ is
 - (i) Non-homologous end-junction
 - (ii) Non-homologous end-joint
 - (iii) Non-homologous end-jump
 - (iv) Non-homologous end-joining
12. Select the odd process
 - (i) Protein folding
 - (ii) Polyadenylation
 - (iii) Proteolytic cleavage
 - (iv) Intein splicing
13. Wobble hypothesis associated with
 - (i) Gene mutation
 - (ii) Genetic code
 - (iii) Gene sequencing
 - (iv) Genome Size
14. mRNA sequence is similar to
 - (i) Anti-sense strand or minus strand
 - (ii) Sense strand or plus strand
 - (iii) Template strand or anti-sense strand
 - (iv) None of the above
15. Specific regions for protein synthesis are found in
 - (i) Ribosome
 - (ii) Endoplasmic reticulum
 - (iii) Nucleus
 - (iv) Mitochondria
16. miRNA interferes _____ thereby affects amount of protein production in a cell
 - (i) tRNA
 - (ii) rRNA
 - (iii) mRNA
 - (iv) All types of RNAs
17. *Cro* gene is associated with *E. coli* replication True/ False
18. *trp* operon is an example of a repressible as well as an inducible system True/ False
19. Alterations in the nucleotide sequences in the intergenic regions are called silent mutations True/ False
20. DNA recombination results in insertion and/ or deletion of nucleotides True/ False

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Course code & Name: BS309 Molecular Biology

06.11.2019

Wednesday

10:31 AM TO 01:00 PM

Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEETS
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Attempt ANY FOUR

20

1. How doubling of genetic material, in the organisms lacking nuclear envelop, takes place? Describe the process.
2. Describe briefly, the processes that leads to the formation of active form a specific type of macromolecule that play crucial role in cells structure and various metabolic activities.
3. Describe structural features of the nucleic acid that makes up the genome of HIV or TMV
4. What are transposable elements? Discussion the association of transposable elements with Zea mays kernels.
5. How a codon containing transcript is synthesized and processed in the organisms where DNA is enclosed in a double membrane structure.

SECTION – II

Q - III Answer the following questions as directed

30

1. Attempt ANY FOUR

20

- (i) Describe various mechanisms of DNA repair with a special focus on double stranded DNA breaks.
- (ii) Who developed the concept of “transforming principle” and how? Discuss any one of the later experiments that proved one of the macromolecules to be the transforming principle.
- (iii) Describe genome organization in nuclear pore containing organisms
- (iv) For which purpose ribonucleo-particles and ribonucleo-acid that resembles cloverleaf shape, comes together in the cell. Discuss their role.
- (v) Discuss gene mutation
- (vi) Discuss regulation of *trp* operon

2. Describe briefly, ANY FIVE

10

- (i) Linking Number, Twist and Writhe
- (ii) C-value Paradox
- (iii) RNA as genetic material
- (iv) Genetic code
- (v) Molecular switch of phage lambda
- (vi) miRNA and their role in gene expression